Student Batch Performance Analyzer

Create user-defined functions in Python to analyze student performance data.

Your functions should:

1. calculate_average(values)
   Return the average of a list of numbers.
2. find_top_performers(data, column, threshold)
   Return students whose score in the given column is greater than or equal to the threshold.
3. find_at_risk_students(data, column, threshold)
   Return students whose score in the given column is below the threshold.
4. batch_summary(data)
   Return summary insights such as:
   - total students
   - average attendance
   - average assignment submission
   - average total consumption
5. rank_batches(batch_data)
   Compare multiple batches and return the best-performing batch.

Sample challenge statement:

You are given student performance data for three batches: b7, b8, and b9.
Write Python user-defined functions to calculate batch-level insights and identify:

- best batch overall
- top-performing students
- at-risk students
- average attendance
- average assignment submission
- average content consumption

Expected output:

Best Batch: b9

Batch Summary:
b7 - Avg Consumption: 46.4%, Avg Attendance: 56.6%
b8 - Avg Consumption: 48.3%, Avg Attendance: 62.3%
b9 - Avg Consumption: 51.7%, Avg Attendance: 69.1%

At-Risk Students:
Students with consumption below 20%

This is a practical data analyst challenge because it tests:
functions, lists, dictionaries, conditions, loops, aggregation, and basic analytics logic.

DataSet(List of Dictionaries)

data = [

{"batch": "b7", "name": "Anil", "consumption": 72, "live": 65, "recording": 70, "assignment_sub": 80, "assignment_score": 75},

{"batch": "b7", "name": "Meena", "consumption": 45, "live": 50, "recording": 55, "assignment_sub": 40, "assignment_score": 35},

{"batch": "b7", "name": "Ravi", "consumption": 18, "live": 20, "recording": 25, "assignment_sub": 10, "assignment_score": 15},

{"batch": "b7", "name": "Pooja", "consumption": 85, "live": 75, "recording": 80, "assignment_sub": 90, "assignment_score": 88},

{"batch": "b7", "name": "Kiran", "consumption": 30, "live": 40, "recording": 45, "assignment_sub": 25, "assignment_score": 28},

{"batch": "b8", "name": "Asha", "consumption": 90, "live": 85, "recording": 88, "assignment_sub": 92, "assignment_score": 90},

{"batch": "b8", "name": "Manoj", "consumption": 55, "live": 60, "recording": 65, "assignment_sub": 50, "assignment_score": 45},

{"batch": "b8", "name": "Deepa", "consumption": 22, "live": 30, "recording": 35, "assignment_sub": 20, "assignment_score": 18},

{"batch": "b8", "name": "Rahul", "consumption": 78, "live": 70, "recording": 75, "assignment_sub": 80, "assignment_score": 76},

{"batch": "b8", "name": "Sneha", "consumption": 10, "live": 15, "recording": 20, "assignment_sub": 5, "assignment_score": 8},

{"batch": "b9", "name": "Vikram", "consumption": 88, "live": 82, "recording": 90, "assignment_sub": 85, "assignment_score": 80},

{"batch": "b9", "name": "Divya", "consumption": 60, "live": 68, "recording": 72, "assignment_sub": 65, "assignment_score": 60},

{"batch": "b9", "name": "Arjun", "consumption": 35, "live": 40, "recording": 50, "assignment_sub": 30, "assignment_score": 25},

{"batch": "b9", "name": "Neha", "consumption": 95, "live": 90, "recording": 95, "assignment_sub": 98, "assignment_score": 96},

{"batch": "b9", "name": "Karthik", "consumption": 15, "live": 20, "recording": 25, "assignment_sub": 10, "assignment_score": 12}

]